

LEARNING METHODS

IN SOCIAL MEDIA

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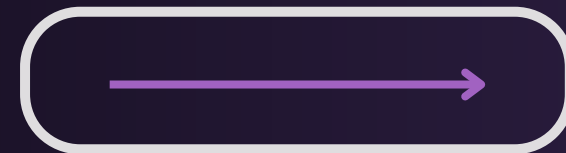
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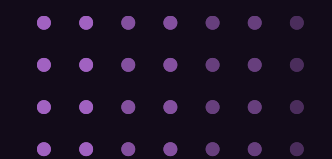
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OVERVIEW

PLATFORM SELECTION

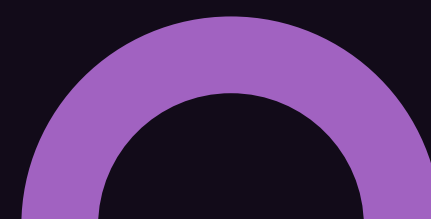
- The case studies include Facebook, TikTok, LinkedIn, and Xiaohongshu
- These platforms represent diverse ecosystems and technical architectures

SUPERVISED EXAMPLES

- Facebook and TikTok mainly apply supervised prediction for ranking
- They rely on labeled behavioral data such as CTR and CVR signals

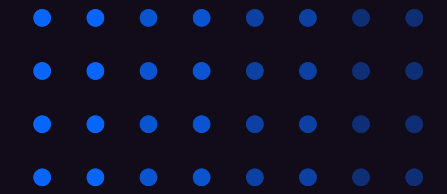
UNSUPERVISED EXAMPLES

- LinkedIn and Xiaohongshu rely heavily on unsupervised clustering and embeddings
- These methods discover communities and user-interest structures



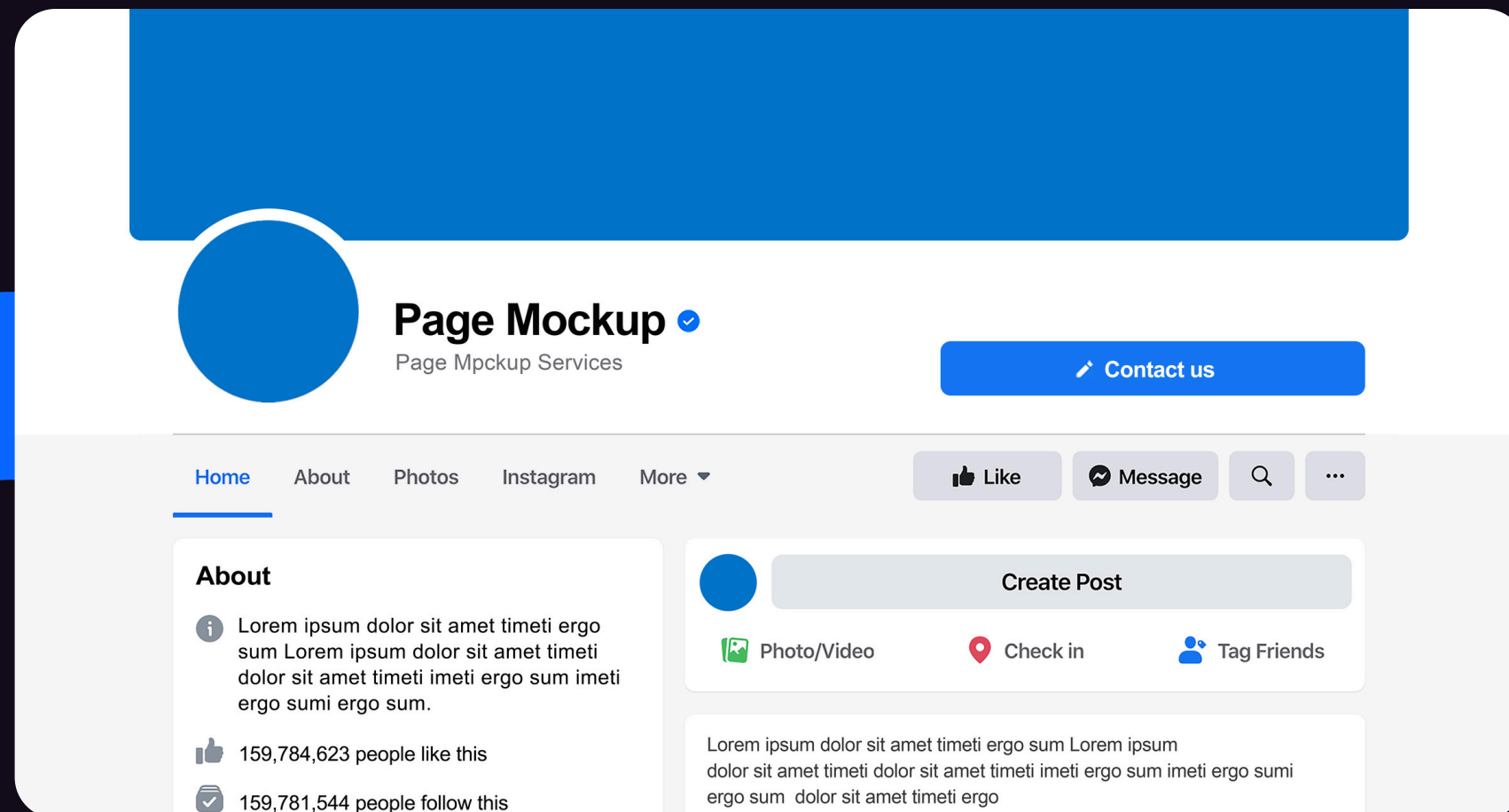


SUPERVISED — FACEBOOK



FACEBOOK:

SYSTEM OVERVIEW



CTR PREDICTION

- Facebook's ad delivery system relies heavily on CTR prediction.
- It uses a hybrid GBDT + Logistic Regression model.

HYBRID MODEL ROLE

- GBDT captures nonlinear feature interactions among diverse signals.
- Logistic Regression produces calibrated CTR probabilities.

IMPACT ON ADS

- These predictions directly determine which ads users see.
- Supervised learning is essential to Facebook's revenue engine.



FACEBOOK: MODEL ARCHITECTURE

GBDT COMPONENT

- GBDT partitions the feature space into meaningful nonlinear regions.
- Each leaf index becomes a transformed categorical feature.

LOGISTIC REGRESSION LAYER

- LR consumes these transformed features for efficient linear scoring.
- This ensures fast real-time predictions.

EVOLUTION OF THE SYSTEM

- Later systems added neural embeddings for stronger representation learning.
- But the supervised framework remains the foundation.

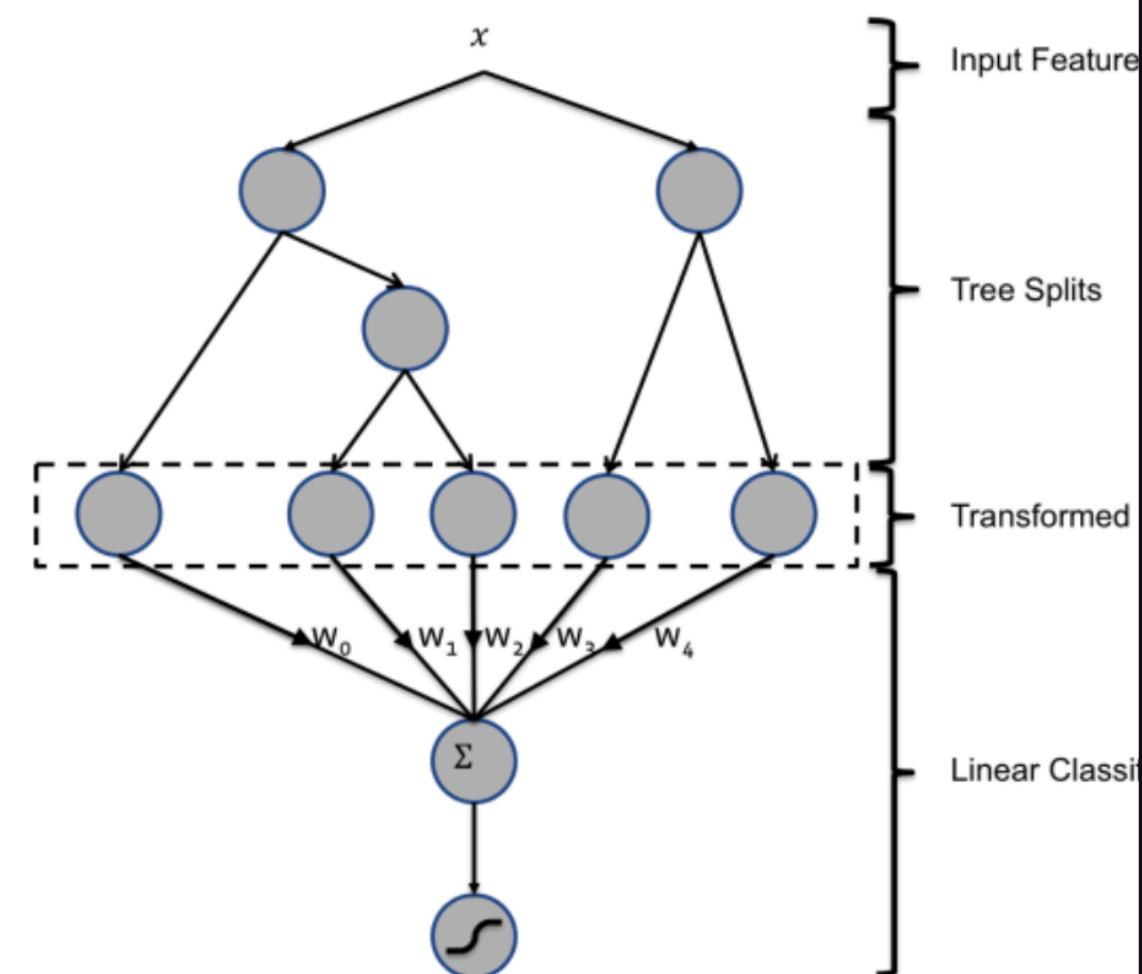


Figure 1: Hybrid model structure. Input features are transformed by means of boosted decision trees. The output of each individual tree is treated as a categorical input feature to a sparse linear classifier. Boosted decision trees prove to be very powerful feature transforms.

Hybrid GBDT + Logistic Regression architecture for CTR



FACEBOOK: WHY IT MATTERS

FAIRNESS CONCERNS

- Research shows ad delivery may reinforce social inequalities.
- Even without protected attributes, historical bias can propagate.

ECONOMIC IMPACT

- Ad targeting shapes opportunities in jobs and housing.
- Supervised models influence which groups receive which ads.

NEED FOR TRANSPARENCY

- Because supervised systems rely on past behavior, bias can amplify.
- Accountability requires understanding their data pipelines.



SUPERVISED — TIKTOK



TIKTOK: RECOMMENDATION SYSTEM



DLRM FRAMEWORK

- TikTok uses Deep Learning Recommendation Models at massive scale.
- These models train on billions of user-video interactions daily.

SUPERVISED NATURE

- Labels come from explicit behaviors like clicks, likes, and watch time.
- Thus the system is fundamentally supervised.

INDUSTRIAL SCALE

- Training requires highly optimized distributed computation.
- Continuous updates keep predictions aligned with user interest shifts.



TIKTOK: MONOLITH ARCHITECTURE

COLLISIONLESS EMBEDDINGS

- The Monolith framework maintains large-scale embedding tables.
- It prevents collisions across massive vocabulary sizes.

FEATURE INTEGRATION

- Both sparse and dense features are incorporated into the model.
- User and content data interact to form predictions.

REAL-TIME UPDATES

- The system processes frequent updates from live user behavior.
- This ensures ranking relevance remains fresh.

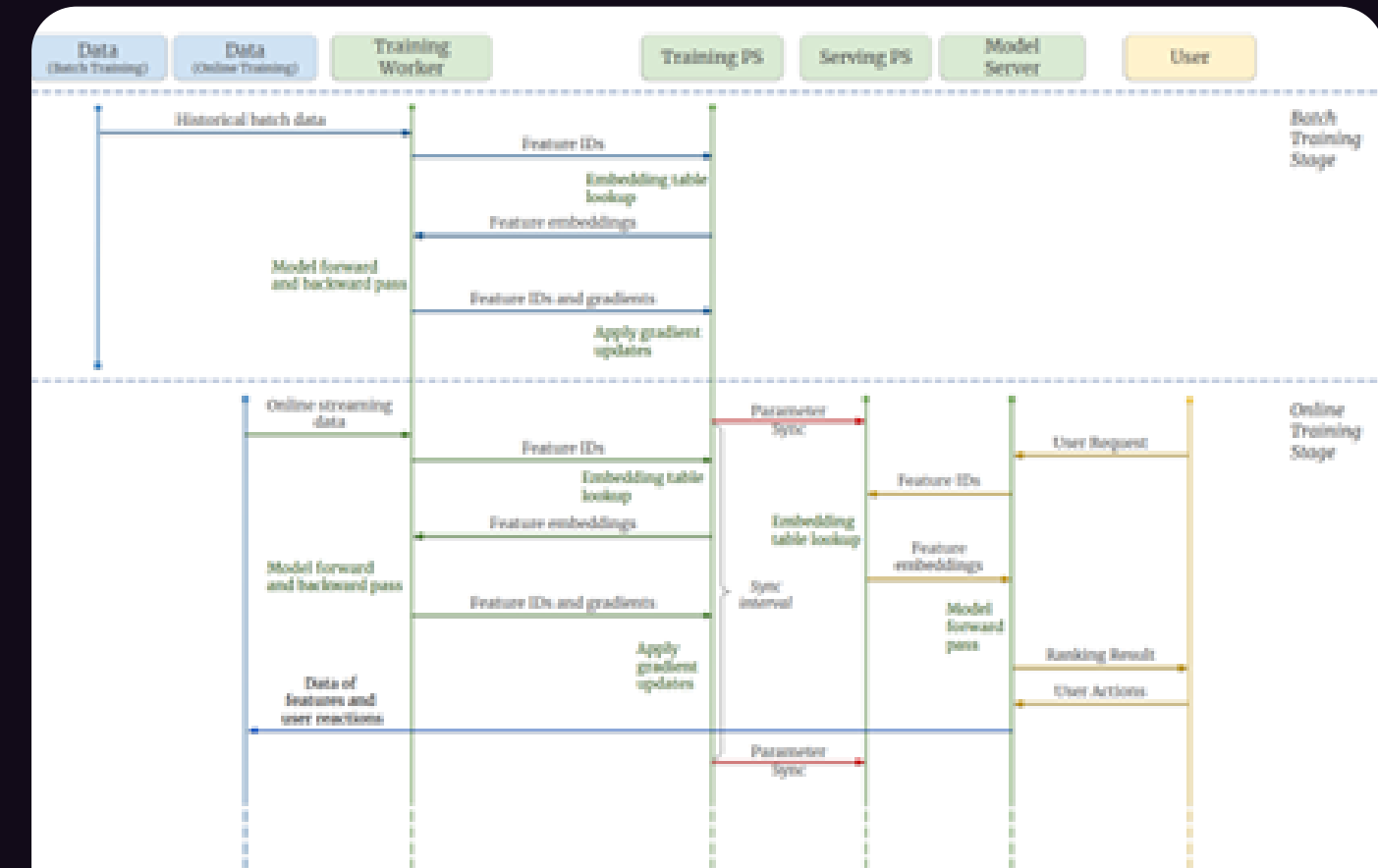


Figure 1: Monolith Online Training Architecture.



TIKTOK:

INNOVATIONS IN SUPERVISED MODELING

ADAPTIVE DOMAIN SCALING (ADS)

- ADS allows one network to serve multiple recommendation domains.
- Selected layers scale for specialization without duplicating full models.

LONGER MODEL

- LONGER introduces self-supervised pretraining on long sequences.
- It then fine-tunes with supervised CTR and CVR labels.

MODEL PERFORMANCE

- These innovations enhance accuracy, personalization, and adaptability.
- TikTok continues to refine its supervised pipeline.



UNSUPERVISED — LINKEDIN

LINKEDIN: GRAPH OVERVIEW



GRAPH REPRESENTATION

- LinkedIn models users, companies, skills, and jobs as nodes.
- Relationships form a massive Economic Graph.

UNSUPERVISED CANDIDATE GENERATION

- “People You May Know” relies on graph-mining techniques.
- Structural proximity is detected using unsupervised algorithms.

ROLE IN RECOMMENDATIONS

- Unsupervised recall generates potential candidates.
- Supervised ranking follows afterward.



LINKEDIN:

GRAPH MINING TECHNIQUES

MULTI-HOP NEIGHBORHOODS

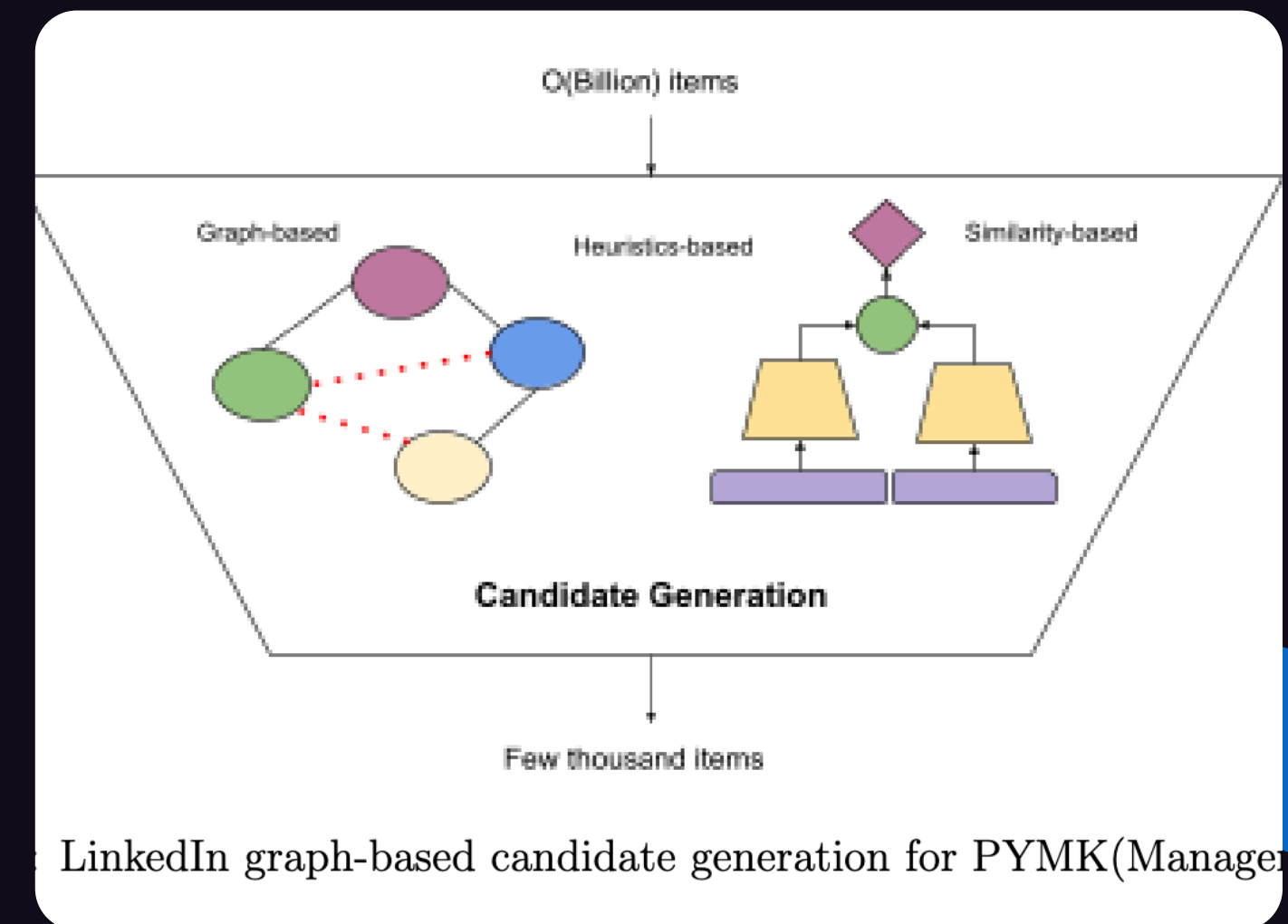
- The system explores multi-hop relationships in the graph.
- This reveals indirect but meaningful user connections.

COMMUNITY DETECTION

- Techniques like Louvain identify overlapping professional communities.
- These clusters reflect shared skills and career paths.

SCALABILITY

- Graph mining enables recall across billions of nodes.
- It reduces the search space before ranking.



LINKEDIN:

WHY UNSUPERVISED WORKS



HIDDEN STRUCTURES

- Unsupervised graph signals reveal latent professional patterns.
- These relations are hard to capture with labeled data alone.

DIVERSITY BENEFITS

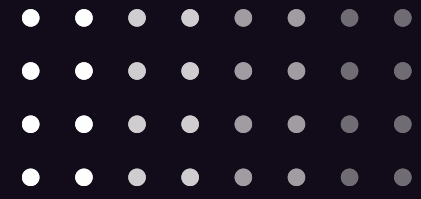
- Clustering helps surface diverse recommendation options.
- Users connect with meaningful but non-obvious profiles.

SYSTEM EFFICIENCY

- Unsupervised recall boosts retrieval quality.
- It strengthens the hybrid recommendation pipeline.



UNSUPERVISED — XIAOHONGSHU



小红书

XIAOHONGSHU: OVERVIEW OF SSD

INTERACTION SEQUENCES

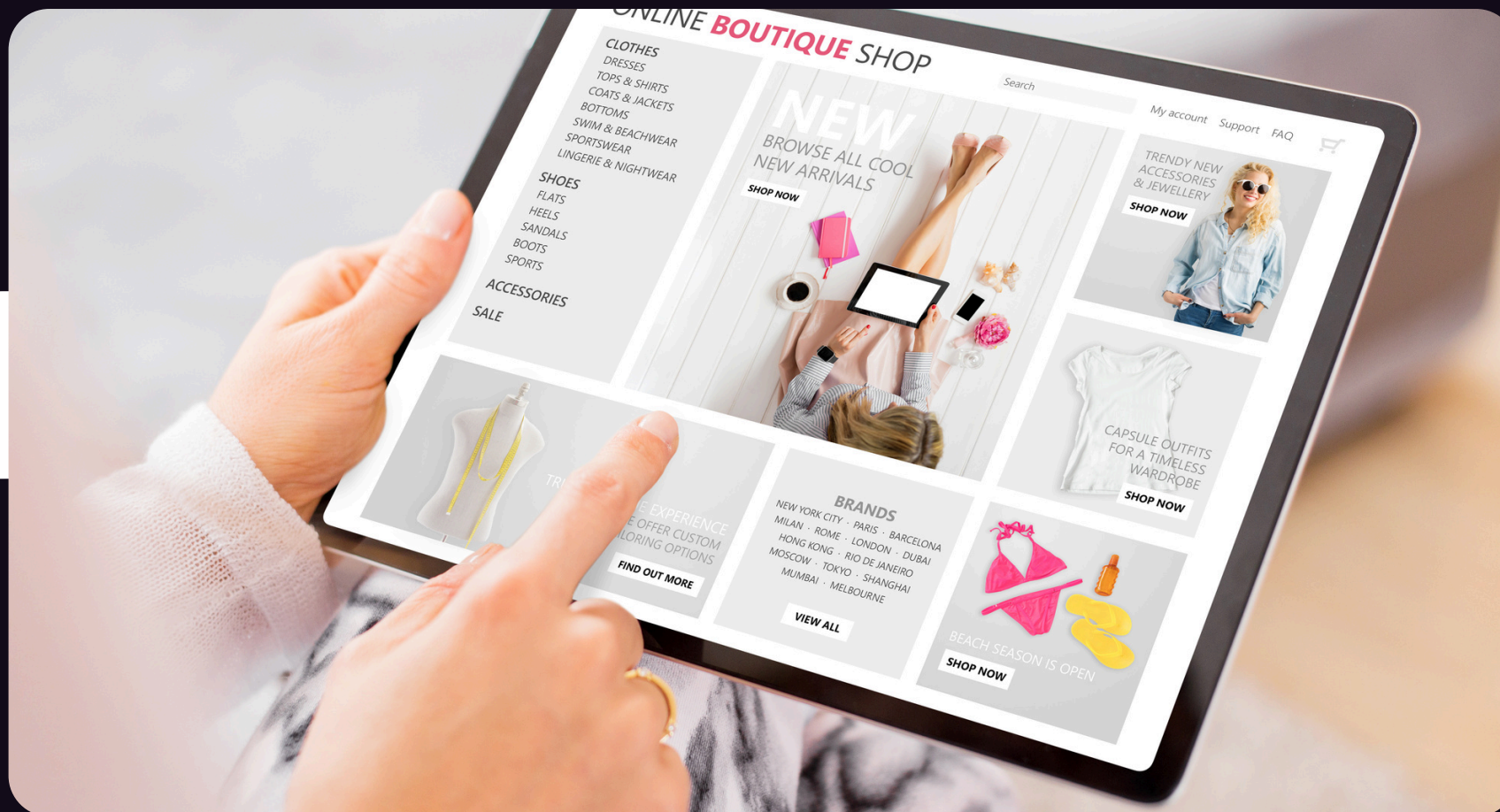
- Xiaohongshu analyzes long sequences of browsing, clicking, and saving.
- These sequences serve as the basis for unsupervised learning.

SSD TECHNIQUE

- Sliding Spectrum Decomposition learns latent preference vectors.
- It captures both stable and shifting user interests.

LABEL-FREE LEARNING

- SSD requires no manually labeled data.
- It naturally clusters users and content.





XIAOHONGSHU: SSD DETAILS

LONG-TERM INTEREST COMPONENT

- SSD identifies stable preferences over long time periods.
- These components reflect consistent user behaviors.

SHORT-TERM COMPONENT

- SSD also captures short-term fluctuations in interest.
- This makes recommendations more adaptive.

EMBEDDING DIVERSITY

- Items sampled from different embedding regions increase feed diversity.
- This improves user experience.



Figure 1: A screenshot of Xiaohongshu. Left: a 2-column list of recommendations that can slide and choose from. Right: a detailed view of an item after a user clicks on it.

小红书



XIAOHONGSHU: SYSTEM IMPACT

DIVERSITY IMPROVEMENT

- SSD reduces repetitive content loops in the feed.
- Users are exposed to more varied content.

COMMUNITY DISCOVERY

- Clustering uncovers interest-based communities.
- These clusters help generate relevant recommendations.

COCOON MITIGATION

- Embedding diversity prevents over-personalization.
- It helps avoid information cocoons.

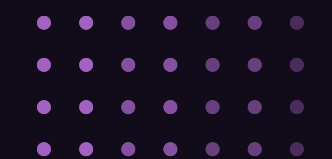


1. HYBRID WORKFLOW OF SOCIAL MEDIA ALGORITHMS

2. ETHICAL RISKS IN ALGORITHMIC PERSONALIZATION

3. HUMAN-CENTERED SOLUTIONS AND KEY TAKEAWAYS





HOW HYBRID SYSTEMS WORK



HYBRID PIPELINE

- Unsupervised models generate large-scale candidate lists.
- Supervised models rank these candidates using CTR/CVR predictions.



FEEDBACK LOOP

- User interactions become new labels for supervised learning.
- They also reshape embeddings for unsupervised models.



BENEFITS

- Supervised learning boosts accuracy and personalization.
- Unsupervised learning increases diversity and scalability.

ETHICAL RISKS



SUPERVISED RISKS

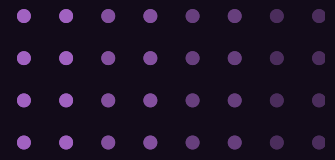
- Models trained on historical data may reinforce social inequalities.
- Facebook ads and TikTok engagement systems can amplify bias and filter bubbles.

UNSUPERVISED RISKS

- Clustering may isolate minority groups or narrow user identities.
- Embedding-based segmentation depends on large-scale behavioral tracking.

SHARED PROBLEMS

- Both methods shape visibility and exposure in unequal ways.
- Transparency and user control remain limited across platforms.



HUMAN-CENTERED SOLUTIONS & TAKEAWAYS

USER-FACING TOOLS

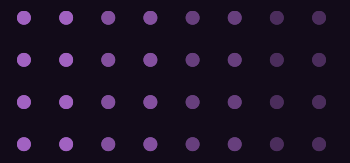
- Dashboards and visualizations help users understand filter bubbles.
- Generative agents and “diversity modes” broaden exposure.

FAIRNESS APPROACHES

- Causal models can reduce reinforcement of biased patterns.
- Exposure-based regularization promotes diverse content.

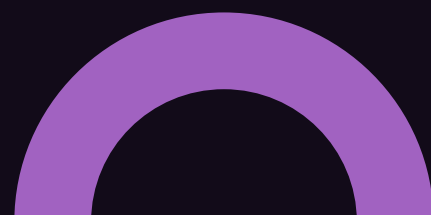
MAIN CONCLUSION

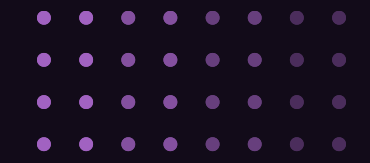
- Supervised and unsupervised methods complement each other in modern platforms.
- Future systems need both technical fixes and human-centered design for responsible use.



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TEAM CONTRIBUTION

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- Group leader led the full project;
- Consolidated the full written report;
- Structured and designed the presentation slides;
- ensured accuracy, coherence, and quality of all sections.

ZHANG YUFEI
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- Co-consolidated the written report;
- Responsible for the TikTok supervised-learning case study
- Conducted comparative research on TikTok recommendation mechanisms to strengthen the technical context of the supervised-learning case study..

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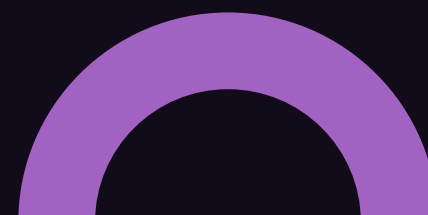
- Edited and produced the final presentation video;
- Responsible for the LinkedIn case study.
- Optimized the video narrative flow and visual pacing, maintaining clarity and engagement.

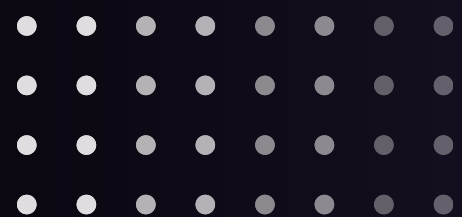
FAN RUITING
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- Responsible for the Facebook case study.
- Reviewed relevant Facebook algorithm literature and integrated practical industry insights to enhance the depth of the supervised-learning analysis.

WU MENGTING
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- Responsible for the Xiaohongshu case study and unsupervised-learning analysis.
- Designed the clustering comparison section for Xiaohongshu, illustrating how different unsupervised methods yield distinct content-grouping patterns.





THANK YOU !

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